STRICT DIAGONAL DOMINANCE DOES NOT ENSURE DIMINISHING NYSTRÖM ERROR REDUCTIONS

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ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: A strictly diagonally dominant positive-definite 4×4 signed matrix has a nonempty-base marginal Nyström error reduction that increases after an additional index is selected. *Certificate:* An exact integer 4×4 matrix, rational shift, four rational principal inverse traces, and the positive violation gap 2165/14833896.

Posed by Mark Fornace (2026); counterexample found by OpenAI Codex (2026-08); formalized by OpenAI Codex; audited by OpenAI Codex; contributed by Suvrit Sra.

For a symmetric positive-definite matrix L and $\gamma > 0$, put $K = (L + \gamma I)^{-1}$. For a nonempty selected index set $\mathcal{I}$, the nuclear Nyström error is

$$F(\mathcal{I}) = \left\| K - K_{:,\mathcal{I}} K_{\mathcal{I},\mathcal{I}}^{-1} K_{\mathcal{I},:} \right\|_* .$$

The residual is positive semidefinite, so its nuclear norm equals its trace. The intended diminishing-error-reduction property is that, for $S \subseteq T$ and $i \notin T$,

$$F(S) - F(S \cup \{i\}) \geq F(T) - F(T \cup \{i\}).$$

This formulation avoids the convention-dependent choice between the words "submodular" and "supermodular" for a decreasing error function. The property is what would license selecting $\mathcal{I}$ greedily; without it, landmark sets are chosen by other means, for instance by determinantal sampling (Li et al., 2016) or through elementary-symmetric-polynomial design objectives (Mariet and Sra, 2017). Colbrook had already resolved Problem 1 in July 2026, proving the SDDM case and giving exact SDD obstructions (Colbrook, 2026). The matrix below was found independently in August 2026; no priority over Colbrook is claimed.

Problem 1 (As posed: Problem 4.6 of Amsel et al. (Amsel et al., 2026), posed there by Mark Fornace. Colbrook (Colbrook, 2026) subsequently completed the answer before this independent witness was found.)**.** Prove or disprove the submodularity of the nuclear Nyström error (14) when L is assumed, in contrast to the above, to be (a) SDDM and positive-definite or (b) SDD and positive-definite.

Problem 1 asks whether the nuclear Nyström error retains diminishing reductions for positive-definite SDDM or SDD matrices (Amsel et al., 2026). Part (a) is true, while part (b) is false (Colbrook, 2026). Here is an independently found exact witness for part (b) with a nonempty base set.

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Theorem 2 ([**Disproof**]: SDD Nyström diminishing returns)**.** *The nuclear Nyström error need not have diminishing error reductions when L is symmetric diagonally dominant and positive definite, even when L is strictly diagonally dominant and the comparison uses a nonempty base set.*

Proof. Let

$$M = \begin{pmatrix} 11 & 3 & -4 & 3 \\ 3 & 12 & 4 & -1 \\ -4 & 4 & 14 & -2 \\ 3 & -1 & -2 & 11 \end{pmatrix}, \qquad \gamma = \frac{1}{2}, \qquad L = M - \frac{1}{2}I,$$

and set $K = (L + \gamma I)^{-1} = M^{-1}$. The diagonal entries of L minus the sums of the absolute values of the corresponding off-diagonal entries are

$$\frac{1}{2}, \quad \frac{7}{2}, \quad \frac{7}{2}, \quad \frac{9}{2}.$$

Thus L is symmetric strictly diagonally dominant with positive diagonal; Gershgorin's theorem gives $L \succ 0$.

For a selected set $\mathcal{I}$ with complement J, the block inverse identity gives

$$K_{J,J} - K_{J,\mathcal{I}} K_{\mathcal{I},\mathcal{I}}^{-1} K_{\mathcal{I},J} = (M_{J,J})^{-1}.$$

All other blocks of the Nyström residual vanish. Consequently

$$F(\mathcal{I}) = \operatorname{Tr}\big((M_{J,J})^{-1}\big).$$

Use one-based indices, take the nonempty base $S = \{2\}$, and compare adding $i = 3$ before and after $j = 4$. Direct exact inversion of the four relevant principal submatrices gives

$$F(\{2\}) = \frac{100}{349}, \qquad F(\{2,3\}) = \frac{11}{56}, \qquad F(\{2,4\}) = \frac{25}{138}, \qquad F(\{2,3,4\}) = \frac{1}{11}.$$

For example, the complement of $\{2\}$ is indexed by $\{1,3,4\}$; its 3×3 principal submatrix has determinant 1396 and the sum of its three diagonal cofactors is 400, yielding $400/1396 = 100/349$.

The two marginal error reductions are therefore

$$F(\{2\}) - F(\{2,3\}) = \frac{1761}{19544}, \qquad F(\{2,4\}) - F(\{2,3,4\}) = \frac{137}{1518}.$$

Their difference in the forbidden direction is

$$\frac{137}{1518} - \frac{1761}{19544} = \frac{2165}{14833896} > 0.$$

Hence selecting index 3 reduces the error more, not less, after index 4 has already been selected. This exactly violates diminishing error reductions. Every displayed quantity is an integer or an exact rational. □

Code and certificate

The case is the directory `counterexamples/sdd-nystrom-diminishing-returns/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/sdd-nystrom-diminishing-returns
python verify.py
```

- Theorem 2 (result `sdd-nystrom-diminishing-returns`): exact arithmetic.
- `artifacts/certificate.json`: Exact matrix, diagonal-dominance margins, principal inverse traces, marginal reductions, and positive violation gap

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by OpenAI Codex (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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